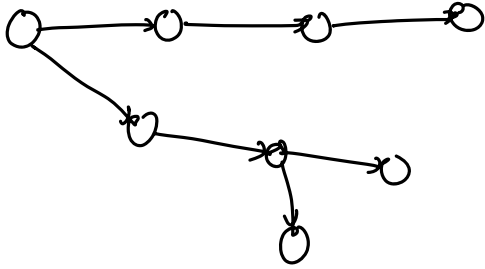


Feb 13, 10:30

Kripke structure:

$K : \langle \text{states}, \text{initstate}, \text{trans}, \text{proposition}, \text{label} \rangle$



$$\text{Trans} = \Sigma (s_i, s_j)$$

$$\text{proposition} = \{ (x \geq 5), (p = 1), \dots \}$$

$$\text{label} : \text{States} \rightarrow 2^{\text{proposition}}$$

$$\text{label}(s_i) = \text{prop}$$

Kripke structure at s satisfies

$$K, s \models \text{AR} [\phi_1, \phi_2]$$

$$= \forall \pi : \left(\forall 0 \leq i < j \ \pi^i \models \phi_1 \right) \rightarrow \left(\pi^j \models \phi_2 \right)$$

$$\neg (AX p) \quad \cong \quad EX \neg p$$

$$\neg (AF p) \quad = \quad EG \neg p$$

$$\neg (AG p) \quad = \quad EF \neg p$$

$$\neg (EG p) \quad = \quad AF \neg p$$

$$\neg (EF p) \quad = \quad AG \neg p$$

semantic equivalence
 $\hookrightarrow$ exam

$$\neg (AU [p, q]) \quad = \quad ER [\neg p, \neg q]$$

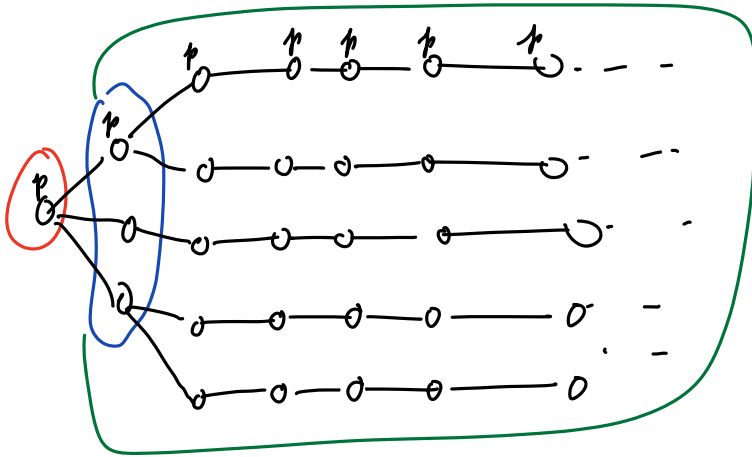
$\top \rightarrow$ top operator, symbol to represent true

$$\textcircled{1} \left[\begin{array}{lcl} A[f \text{ W } g] & = & \neg E[\neg g \cup (\neg f \wedge \neg g)] \\ E[f \cup g] & = & \neg A[\neg g \text{ W } (\neg f \wedge \neg g)] \end{array} \right.$$

$$\textcircled{2} \left[\begin{array}{lcl} AG f & = & f \wedge AX AG f \end{array} \right.$$

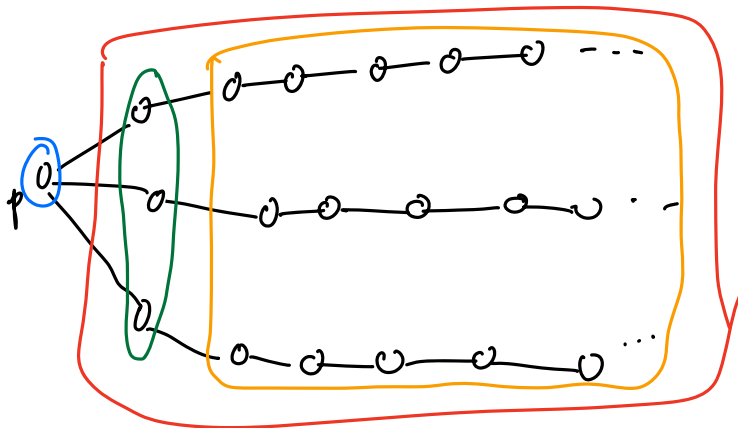
conjunction

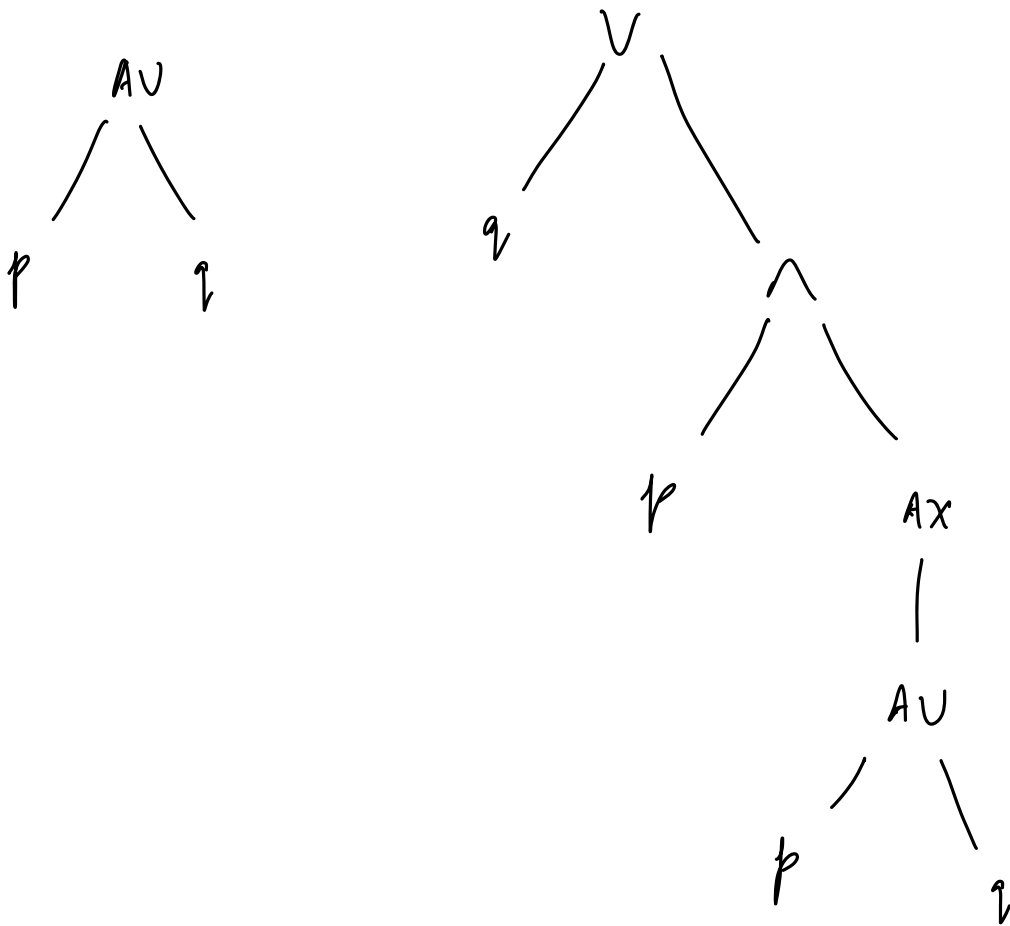
$$AG\ p = \textcircled{p} \wedge \textcircled{AX\ AG\ p}$$



"axiomatic"

$$AF\ p = \textcircled{p} \vee \textcircled{AX\ AF\ p}$$

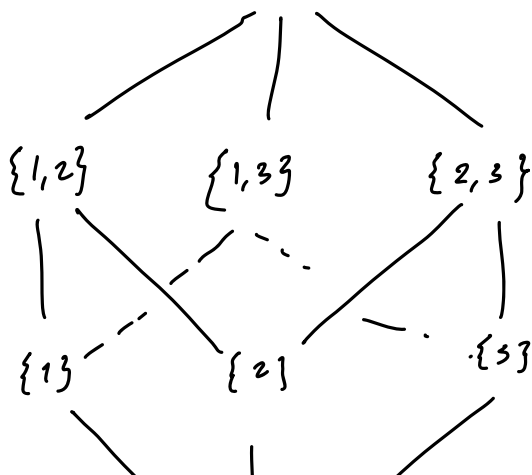




Feb 17, 11:45

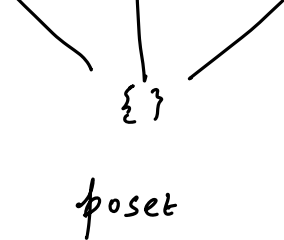
① Taking union of set

Lattice $S = \{1, 2, 3\}$



Lattice - power set of

$$2^S$$



fixed point characterisation $f(x) = x$

$$f: X \rightarrow X$$

to find a fixed point, f should be "monotonic,"

domain x should be partially ordered.

x : partially ordered

new material in drive $\rightarrow$ restrict to just few dopters
(CTL only)

$$\underbrace{AG p}_x = \underbrace{p \wedge AX}_{f} \underbrace{AG p}_x$$

$$f(z) = p \wedge AX(z)$$

if we get $f(z) = z$, we have found the solution.

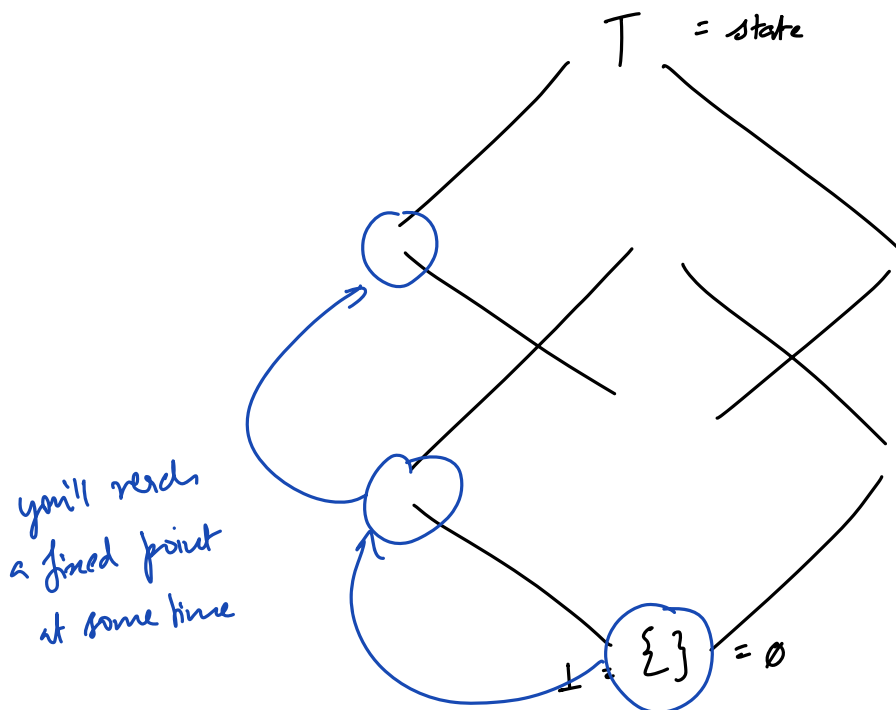
$$\underbrace{AF q}_z = \underbrace{q \vee AX AF q}_f$$

$$f(z) = q \vee AX(z)$$

Model checking

$K \models \phi?$

$\hookrightarrow \forall s \in \text{init states}$



greatest fixed point
 $gfp(t) \quad \forall f$

lowest fixed point
 $lfp(t) \quad \mu f$

$$[[AG \phi]] = \underbrace{\nu z. f_{(AG, \phi)}(z)}_{\text{greatest fixed point}} \text{ where } f_{(AG, \phi)}(z) = [[\phi]] \cap \text{pred}_\forall(z)$$